# MAT627 Chapter 2.5 Programming Assignment

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We approximate

$$\int_a^b f(x)\,dx$$

with the composite trapezoid rule

$$T_N(f) = \frac{b-a}{2N}\left[f(x_0) + 2f(x_1) + \cdots + 2f(x_{N-1}) + f(x_N)\right],$$
$$x_i = a + i\frac{b-a}{N}.$$

Equivalently, $T_N(f)$ is the exact integral of the piecewise linear interpolant on the uniform mesh.

## 1 Part 1

Composite trapezoid rule for $f(x) = x^2 e^{-x}$ on $[1, 3]$.

Table 1: Composite trapezoid-rule results on $[1, 3]$.

| $N$ | $h$ | $T_N(f)$ | $E_h = \lvert T_N(f) - I(f)\rvert$ | Rate |
|---|---|---|---|---|
| 2 | 1 | $9.493227 \times 10^{-1}$ | $4.369438 \times 10^{-2}$ | – |
| 4 | 0.5 | $9.821984 \times 10^{-1}$ | $1.081866 \times 10^{-2}$ | 2.013926 |
| 8 | 0.25 | $9.903203 \times 10^{-1}$ | $2.696735 \times 10^{-3}$ | 2.004236 |
| 16 | 0.125 | $9.923434 \times 10^{-1}$ | $6.736653 \times 10^{-4}$ | 2.001110 |
| 32 | 0.0625 | $9.928487 \times 10^{-1}$ | $1.683836 \times 10^{-4}$ | 2.000281 |

## 2 Part 2

Composite trapezoid rule for $f(x) = x^2 e^{-x}$ on $[0, 2]$.

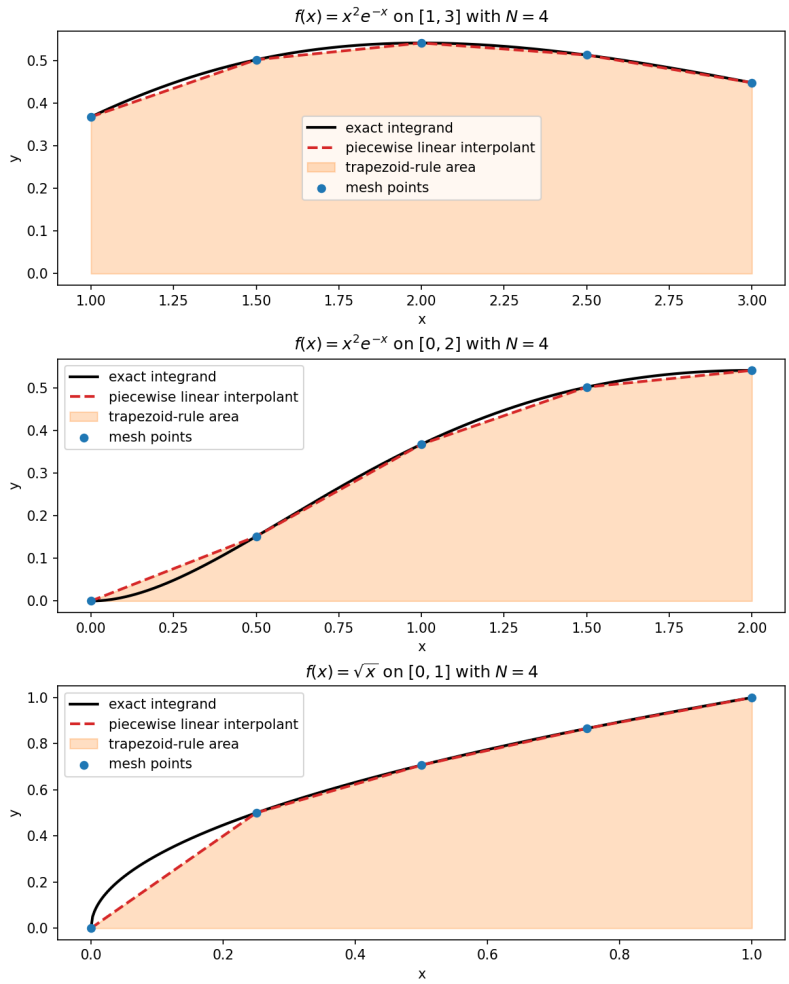


Figure 1: Representative trapezoid-rule geometry using the piecewise linear interpolant with $N = 4$.

Table 2: Composite trapezoid-rule results on $[0, 2]$.

| $N$ | $h$ | $T_N(f)$ | $E_h = \lvert T_N(f) - I(f)\rvert$ | Rate |
|---|---|---|---|---|
| 2 | 1 | $6.385500 \times 10^{-1}$ | $8.097160 \times 10^{-3}$ | – |
| 4 | 0.5 | $6.461128 \times 10^{-1}$ | $5.344012 \times 10^{-4}$ | 3.921421 |
| 8 | 0.25 | $6.466133 \times 10^{-1}$ | $3.386396 \times 10^{-5}$ | 3.980100 |
| 16 | 0.125 | $6.466450 \times 10^{-1}$ | $2.123833 \times 10^{-6}$ | 3.995009 |
| 32 | 0.0625 | $6.466470 \times 10^{-1}$ | $1.328545 \times 10^{-7}$ | 3.998751 |

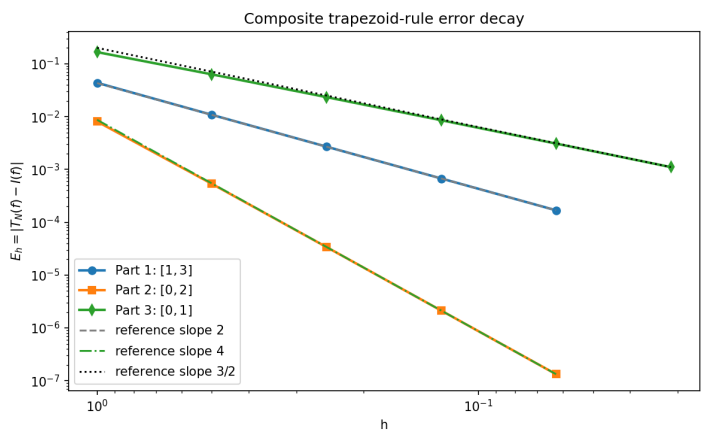


Figure 2: Error decay for the three test problems.

# 3 Part 3

Composite trapezoid rule for $f(x) = \sqrt{x}$ on $[0, 1]$.

Table 3: Composite trapezoid-rule results on $[0, 1]$.

| $N$ | $h$ | $T_N(f)$ | $E_h = \lvert T_N(f) - I(f)\rvert$ | Rate |
|---|---|---|---|---|
| 1 | 1 | $5.000000 \times 10^{-1}$ | $1.666667 \times 10^{-1}$ | – |
| 2 | 0.5 | $6.035534 \times 10^{-1}$ | $6.311328 \times 10^{-2}$ | 1.400950 |
| 4 | 0.25 | $6.432830 \times 10^{-1}$ | $2.338362 \times 10^{-2}$ | 1.432445 |
| 8 | 0.125 | $6.581302 \times 10^{-1}$ | $8.536445 \times 10^{-3}$ | 1.453791 |
| 16 | 0.0625 | $6.635812 \times 10^{-1}$ | $3.085470 \times 10^{-3}$ | 1.468145 |
| 32 | 0.03125 | $6.655589 \times 10^{-1}$ | $1.107730 \times 10^{-3}$ | 1.477883 |

# 4 Discussion

For Parts 1 and 2, the integrand is

$$f(x) = x^2 e^{-x}, \qquad f''(x) = (x^2 - 4x + 2)e^{-x},$$

so $f''$ is continuous and bounded on both closed intervals. Therefore Theorem 2.3 guarantees at least second-order convergence.

In Part 1, the observed rate approaches 2, which matches the theorem directly.

In Part 2, the observed rate is closer to 4. This does not contradict the theorem; it means the leading $O(h^2)$ term vanishes. Indeed,

$$f'(x) = (2x - x^2)e^{-x},$$

so

$$f'(0) = f'(2) = 0.$$

The leading trapezoid-rule error term is proportional to $f'(b) - f'(a)$, so on $[0, 2]$ it cancels and the next nonzero term gives fourth-order behavior.

For Part 3, the integrand is

$$f(x) = \sqrt{x}, \qquad f''(x) = -\frac{1}{4x^{3/2}},$$

which is singular at $x = 0$. Theorem 2.3 does not apply on $[0, 1]$. On the first subinterval $[0, h]$, the trapezoid rule uses the line through $(0, 0)$ and $(h, \sqrt{h})$, so

$$T_{[0,h]} = \frac{h}{2}\sqrt{h} = \frac{1}{2}h^{3/2}, \qquad \int_0^h \sqrt{x}\, dx = \frac{2}{3}h^{3/2}.$$

Hence the first cell alone contributes an error of

$$\frac{2}{3}h^{3/2} - \frac{1}{2}h^{3/2} = \frac{1}{6}h^{3/2},$$

which explains why the overall convergence rate is approximately $3/2$ rather than 2.